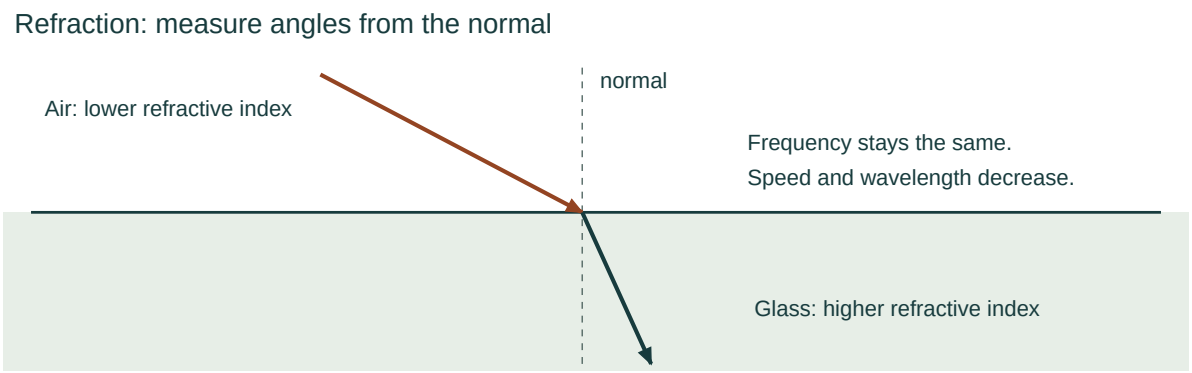
03

Optics and Waves

A field guide to understanding, working and checking.



The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

12 lessons / 2 practice tests / Worked solutions

English edition / November 2026

How to use this handout

1. Read the concept and check the symbol meanings before choosing a formula.
2. Cover the worked steps and reproduce the reasoning yourself.
3. Attempt the quick check and the two subject tests without notes.
4. Mark your work; record whether the error was concept, method, algebra, units or reading. Retry after a delay.

Your study route

1. Wave language and simple harmonic motion
2. Read a travelling wave without deriving it
3. Wavefronts, spreading and group velocity
4. Superposition, standing waves and beats
5. Light, photons and the electromagnetic picture
6. Reflection, refraction and total internal reflection
7. Mirrors, lenses and image signs
8. Optical systems, prisms and instruments
9. Interference and diffraction
10. Polarization and Malus law
11. Scattering, apparent depth and lens power
12. Recognize advanced optics without long derivations

LESSON 01 / 12

Wave language and simple harmonic motion

After this lesson, you can

- Convert an oscillation count into frequency and period.
- Relate displacement, speed and acceleration at key SHM positions.

Start with: One oscillation is one complete cycle; reciprocal quantities; displacement from equilibrium.

THEORY

A wave carries a disturbance and energy. Mechanical waves need a medium; electromagnetic waves can travel through vacuum. Transverse displacement is perpendicular to travel; longitudinal displacement is parallel. Simple harmonic motion has a restoring acceleration proportional and opposite to displacement.

RULE

$$T = 1/f$$

$$f = N/t$$

$$a = -\omega^2 x$$

N number of complete cycles (dimensionless)

t total elapsed time (s)

f cycles per second (Hz)

T time for one cycle (s)

ω angular frequency (rad/s)

A method that earns marks

1. Count complete cycles rather than individual crossings of equilibrium.
2. Divide cycle count by total time for frequency.
3. Find period as total time per cycle or the reciprocal of frequency.
4. For SHM compare the point with equilibrium and use restoring acceleration toward equilibrium.

WORKED EXAMPLE

If 20 oscillations take 5 s, frequency is 4 Hz and period is 0.25 s. At an extreme of SHM, speed is zero and acceleration magnitude is maximum.

1. Twenty complete oscillations in 5 s means 20/5 = 4 cycles per second, so f = 4 Hz.
2. One cycle takes 5/20 = 0.25 s, giving T = 0.25 s.
3. At an extreme the object must turn around, so its instantaneous speed is zero.
4. Its displacement magnitude is largest there; restoring acceleration therefore has maximum magnitude and points toward equilibrium.

Common mistake
The direction of particle motion is not necessarily the direction of wave travel.

Exam habit: Write f = N/t and T = t/N side by side to prevent interchanging hertz and seconds.

SELF-CHECK

At SHM equilibrium is acceleration magnitude maximum?

Answer: No. Displacement and restoring acceleration are zero; speed is maximum.

LESSON 02 / 12

Read a travelling wave without deriving it

After this lesson, you can

- Read amplitude, wavelength and frequency from a travelling-wave expression.
- Use v = fλ and distinguish pattern motion from particle motion.

Start with: Sine-wave shape; π; coefficients of x and t; SI units.

THEORY

Amplitude is maximum displacement, wavelength is crest-to-crest separation, and frequency counts cycles per second. Wave speed is the speed of the pattern, not the oscillating particle. In a sine wave, the coefficients of position and time reveal wavelength and period.

RULE

$$v = f\lambda$$

$$k = 2\pi/\lambda$$

$$\omega = 2\pi f$$

$$y = A\sin(kx - \omega t + \phi)$$

A amplitude or maximum displacement (m)

k wave number (rad/m)

ω angular frequency (rad/s)

λ wavelength (m)

f frequency (Hz)

When this applies

The example takes x and y in metres and t in seconds.

A method that earns marks

1. Match the wave to y = A sin(kx - ωt + ϕ).

2. Read A directly, then divide 2π by k to obtain wavelength.
3. Divide ω by 2π to obtain frequency.
4. Multiply frequency by wavelength for pattern speed and check the units.

WORKED EXAMPLE

For $y = 0.02\sin(4\pi x - 20\pi t)$, amplitude is 0.02 m, wavelength is 0.50 m, frequency is 10 Hz and speed is 5 m/s.

1. Compare y = 0.02 sin(4πx - 20πt) with the standard form: A = 0.02 m, k = 4π rad/m and ω = 20π rad/s.
2. Wavelength is $2\pi/(4\pi)$ = 0.50 m.
3. Frequency is $20\pi/(2\pi)$ = 10 Hz.
4. Pattern speed is 10 times 0.50 = 5 m/s; this is not the maximum transverse speed of an individual particle.

Common mistake
The coefficient of time is angular frequency, not ordinary frequency.

Exam habit: Identify coefficients before calculating; treating k as wavelength or ω as ordinary frequency loses the factor 2π.

SELF-CHECK

Is the coefficient 20π in the example the frequency in hertz?

Answer: No. It is angular frequency; divide it by 2π to get 10 Hz.

Phase means position within a cycle

A phase difference tells whether two oscillations are at the same or different stages of a cycle.

- One full cycle is 2π radians. Half a cycle is π radians.
- At a fixed time, points one wavelength apart are in phase. Points half a wavelength apart are opposite in phase.
- At one place, a full period brings the oscillation back to the same phase.
- These formulas give the size of the phase difference. Its sign depends on the order of comparison and the travelling-wave convention.

$$\Delta\phi = 2\pi\frac{\Delta x}{\lambda}$$ (phase-difference magnitude at one time)

$$\Delta\phi = 2\pi\frac{\Delta t}{T}$$ (phase-change magnitude at one place)

Two points on a sinusoidal wave are separated by $\lambda/4$. Find their phase-difference magnitude at the same time.

1. The separation is one quarter of a full spatial cycle.
2. Therefore $\Delta\phi = 2\pi \times (1/4) = \pi/2$.

$\pi/2$ radians, or $90°$. They are neither in phase nor exactly opposite in phase.

LESSON 03 / 12

Wavefronts, spreading and group velocity

After this lesson, you can

- Predict intensity and amplitude changes for spherical spreading.
- Distinguish wavefront, ray, phase velocity and group velocity.

Start with: Surface area grows with radius squared; square roots; amplitude versus intensity.

THEORY

A ray indicates propagation direction and is normal to a wavefront. Plane wavefronts are flat, spherical wavefronts spread from a point source, and cylindrical wavefronts spread around a line source. Phase velocity tracks a crest; group velocity tracks the envelope of a narrow wave packet.

RULE

$$I \propto A^2$$

$$I_{\text{spherical}} \propto 1/r^2$$

I energy-transfer rate per unit area (W/m^2)

A wave amplitude (depends on disturbance)

r distance from point source (m)

When this applies

The spreading example assumes an isotropic point source, no absorption and unchanged medium properties.

A method that earns marks

1. Identify whether the source produces plane, spherical or cylindrical spreading.
2. For an isotropic point source use intensity proportional to 1/r^2.
3. Convert an intensity ratio to an amplitude ratio by taking its square root.

4. In description questions, identify a crest for phase velocity and the packet envelope for group velocity.

WORKED EXAMPLE

For an isotropic point source without absorption, doubling distance reduces intensity to one quarter and amplitude to one half.

1. The same source power crosses a sphere around the source at each distance.
2. Doubling radius makes the sphere area four times as large.
3. The available power per square metre is therefore one quarter as large.
4. Because intensity is proportional to amplitude squared, the amplitude ratio is $\sqrt{(1/4)} = 1/2$.

Common mistake
The ideal plane-wave model does not have spherical inverse-square spreading.

Exam habit: Do not apply the inverse-square intensity rule automatically to a plane wave or line source.

SELF-CHECK

An intensity of 18 W/m^2 is measured at r. What is it at 3r?

Answer: 2 W/m^2: divide by $3^2 = 9$.

LESSON 04 / 12

Superposition, standing waves and beats

After this lesson, you can

- Calculate beat frequency from two nearby frequencies.
- Connect fixed-string harmonics with allowed wavelengths.

Start with: Adding displacements; wavelength; frequency; fixed endpoints.

THEORY

Overlapping disturbances add algebraically. In-phase equal waves reinforce; opposite-phase equal waves cancel. A standing wave forms when matching waves travel in opposite directions. Nodes stay at zero displacement. Beats are slow changes in loudness from nearby frequencies.

RULE

$$f_{\text{beat}} = |f_1 - f_2|$$

$$\lambda_n = 2L/n \text{ (string fixed at both ends)}$$

f_{beat} number of loudness cycles per second (Hz)

L vibrating string length (m)

n harmonic number, 1, 2, 3, … (dimensionless)

λ_n wavelength of harmonic n (m)

A method that earns marks

1. For superposition add instantaneous signed displacements at the same point.
2. For beats take the positive difference of the two nearby frequencies.
3. For a string fixed at both ends draw nodes at both ends.
4. Fit n half-wavelengths into the length, then use wave speed to find each frequency.

WORKED EXAMPLE

Two tones at 256 and 260 Hz produce 4 beats per second. A fixed string in its fundamental mode contains one half-wavelength.

1. The tones have frequencies 256 Hz and 260 Hz.
2. Their frequency difference is 4 Hz, so there are four loudness beats each second.
3. For the separate string example, the fundamental has one loop between two end nodes.
4. That loop occupies half a wavelength, so $\lambda 1 = 2L$ and $f1 = v/(2L)$.

Common mistake
A node is not a point where two travelling waves are individually absent.

Exam habit: A beat frequency is a difference, not the average of the two tone frequencies.

SELF-CHECK

For a fixed string, how does the second-harmonic frequency compare with the fundamental?

Answer: It is twice the fundamental frequency in the ideal uniform-string model.

LESSON 05 / 12

Light, photons and the electromagnetic picture

After this lesson, you can

- Compare photon energy using frequency or vacuum wavelength.
- Distinguish beam brightness from energy per photon.

Start with: Direct and inverse proportionality; wave frequency; energy transfer.

THEORY

Light behaves as an electromagnetic wave in propagation and exchanges energy in photons. Photon energy increases with frequency. In the photoelectric effect, emission requires enough energy per photon; increasing brightness alone cannot overcome a frequency below threshold.

RULE

$$E_\gamma = hf = hc/\lambda_0$$

$$\lambda_0 = \text{vacuum wavelength}$$

$$c = 3.00 \times 10^8\ \text{m/s}$$

E_γ energy of one photon (J)

h Planck constant (J s)

f light frequency (Hz)

λ vacuum wavelength (m)

A method that earns marks

1. Decide whether the question asks about one photon or the whole beam.
2. Use Eγ proportional to frequency, or inversely proportional to vacuum wavelength.
3. For brightness comparisons separate photon energy from the number arriving each second.
4. For photoelectric emission compare energy per photon with the threshold requirement.

WORKED EXAMPLE

Blue light has shorter wavelength than red light and therefore higher photon energy. A brighter red beam need not have more energetic photons.

1. Blue light has a shorter vacuum wavelength than red light.

2. Since $E\gamma = hc/\lambda$, the smaller wavelength corresponds to a larger energy per photon.
3. Increasing red-beam brightness at the same colour increases the energy arriving per second, usually through more photons.
4. Each red photon still has the same lower energy; brightness alone does not change its frequency.

Common mistake
Higher intensity does not mean higher energy per photon. In E = hc/lambda, use the vacuum wavelength. At an ordinary stationary boundary, frequency and photon energy stay the same even though wavelength in the medium changes.

Exam habit: Use the phrase energy per photon when discussing threshold effects; total beam energy does not answer the same question.

SELF-CHECK

If light A has twice light B frequency, how do photon energy and vacuum wavelength compare?

Answer: A has twice the photon energy and half the vacuum wavelength.

LESSON 06 / 12

Reflection, refraction and total internal reflection

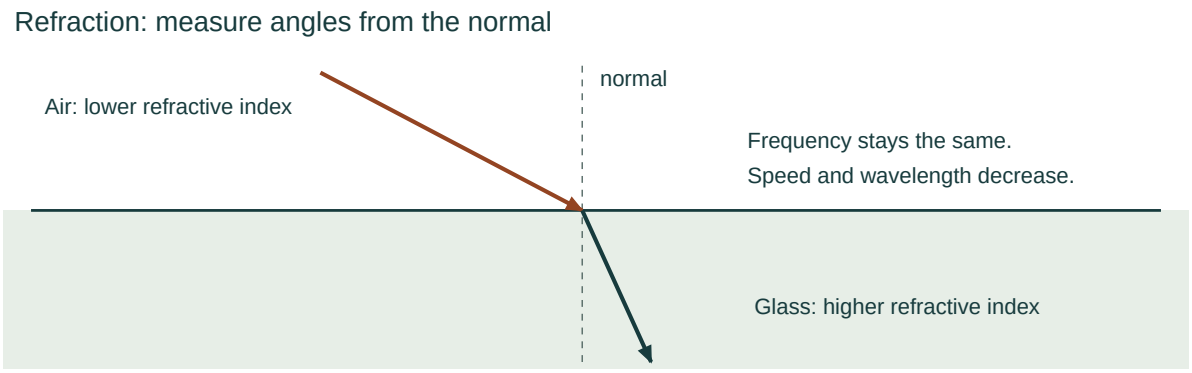
After this lesson, you can

- Track speed, frequency and wavelength across a refracting boundary.
- Recognize the two conditions for total internal reflection.

Start with: Normal to a surface; sine; reciprocal refractive index.

THEORY

Angles of incidence, reflection and refraction are measured from the normal. At a stationary interface frequency stays the same, while speed and wavelength may change. When light enters a higher refractive index at a nonzero angle, the ray bends toward the normal. Total internal reflection needs travel from higher to lower index and an incidence angle greater than the critical angle.



A ray entering glass from air bends toward the normal at oblique incidence. The ray sketch is qualitative; frequency is unchanged while speed and wavelength decrease.

RULE

$$\theta_i = \theta_r$$

$$n = c/v$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$\sin\theta_c = n_2/n_1$$

- n refractive index (dimensionless)
- c vacuum light speed (m/s)
- v light speed in medium (m/s)
- θ_c critical incidence angle (degrees or radians)

When this applies

The boundary is stationary and the media are treated as uniform; air index is approximated as 1.

A method that earns marks

1. Draw the surface normal and measure all ray angles from it.
2. Use v = c/n to compare speeds; frequency stays fixed at a stationary boundary.
3. Use wavelength = speed/frequency to compare wavelengths.
4. For total internal reflection check higher-to-lower index travel, then compare incidence with the critical angle.

WORKED EXAMPLE

From air into glass with index 1.5, light slows to $2 \times 10^8\ \mathrm{m/s}$. Its frequency stays fixed and its wavelength decreases.

1. Air is approximated by n = 1 and glass has n = 1.5.
2. In glass, speed is $3.00 \times 10^8 / 1.5 = 2.00 \times 10^8$ m/s.
3. The frequency is set by the source and remains the same across the boundary.
4. Because speed drops while frequency stays fixed, wavelength in glass is two thirds of its air value.

Common mistake
The critical angle is an incidence angle in the higher-index medium.

Exam habit: At the critical angle the refracted ray runs along the boundary; total internal reflection occurs above that angle.

SELF-CHECK

Can light travelling from air into glass undergo total internal reflection at that interface?

Answer: No. Total internal reflection requires travel from higher to lower refractive index.

LESSON 07 / 12

Mirrors, lenses and image signs

After this lesson, you can

- Use the stated image-distance sign convention consistently.
- Infer image reality, orientation and relative size from di and m.

Start with: Reciprocals; positive and negative numbers; a simple ray sketch.

THEORY

A real image is formed where rays actually meet and can be projected on a screen. A virtual image is where rays appear to originate. Use one sign convention consistently: real object distance positive, real image distance positive; converging focal length positive and diverging focal length negative.

RULE

$$1/f = 1/d_o + 1/d_i$$

$$m = -d_i/d_o$$

f signed focal length (cm or m consistently)

d_o object distance (same length unit as f)

d_i signed image distance (same length unit as f)

m transverse magnification (dimensionless)

When this applies

Use the lesson convention: real object and real image distances positive; converging focal length positive.

A method that earns marks

1. Write the sign convention and assign f and do before substitution.
2. Rearrange the lens equation to isolate 1/di and then invert.
3. Calculate m = -di/do.
4. Use di sign for real or virtual, m sign for orientation, and magnitude of m for size.

WORKED EXAMPLE

For a converging lens with focal length 10 cm and object distance 30 cm, image distance is 15 cm and magnification is -0.5: real, inverted, reduced.

1. A converging lens has f = +10 cm and the real object has do = +30 cm.
2. Then 1/di = 1/10 - 1/30 = 1/15 per cm, so di = +15 cm.
3. Magnification is -15/30 = -0.5.
4. Positive image distance means real; negative magnification means inverted; magnitude 0.5 means half the object height.

Common mistake
Do not call every image behind a lens virtual; interpret actual ray convergence and your convention.

Exam habit: After subtraction, invert 1/di to get di; include all three requested image descriptions.

SELF-CHECK

What does m = +2 tell you about an image?

Answer: It is upright and twice the object height; the sign convention and lens setup determine its image position.

LESSON 08 / 12

Optical systems, prisms and instruments

After this lesson, you can

- Explain the different roles of objective and eyepiece.
- Calculate telescope angular magnification for normal adjustment.

Start with: Focal length; image formation; ratios of lengths.

THEORY

A prism refracts light at two surfaces; dispersion separates colours because refractive index depends on wavelength. In a multi-lens system, the first image becomes the object for the next lens. A microscope and a telescope both use an objective and eyepiece, but they serve different object distances.

RULE

$$m_{\text{total}} = m_1 m_2$$

$$|M_{\text{telescope}}| = f_o / f_e \text{ (normal adjustment)}$$

f_o objective focal length (cm or m consistently)

f_e eyepiece focal length (same length unit as fo)

M angular magnification (dimensionless)

m_1, m_2 transverse magnifications of successive stages (dimensionless)

When this applies

The telescope is in normal adjustment, giving the final image at infinity.

A method that earns marks

1. Identify the instrument and whether it views a distant or nearby object.
2. For a normal-adjustment astronomical telescope divide objective focal length by eyepiece focal length for magnification magnitude.
3. State image orientation separately.
4. For a general multi-lens system locate the first image before using it as the next object.

WORKED EXAMPLE

An astronomical telescope with objective focal length 80 cm and eyepiece 4 cm has angular magnification magnitude 20. Its normal image is inverted.

1. The telescope objective focal length is 80 cm and the eyepiece focal length is 4 cm.
2. Their ratio is 80/4 = 20, so the angular magnification magnitude is 20.
3. An astronomical telescope with this usual two-converging-lens arrangement gives an inverted image.
4. The ratio needs no length conversion because both focal lengths use the same unit.

Common mistake
Linear image magnification and angular magnification are different quantities.

Exam habit: Use the telescope focal-length ratio only for the stated telescope model, not as a universal magnification formula.

SELF-CHECK

Keeping the objective fixed, does doubling the eyepiece focal length increase or decrease telescope magnification?

Answer: It halves the magnification magnitude.

LESSON 09 / 12

Interference and diffraction

After this lesson, you can

- Predict how slit geometry changes an interference or diffraction pattern.
- Distinguish double-slit bright fringes from single-slit dark fringes.

Start with: Wavelength; coherent sources; nanometres and millimetres; small angles.

THEORY

Interference occurs when overlapping waves combine. A stable visible pattern needs a fixed phase relationship (coherence). Diffraction is spreading associated with an opening or obstacle. Narrower slits spread light more. Young double-slit bright fringes are evenly spaced when the angles are small.

RULE

$$\Delta y = \lambda L/d$$

$$d\sin\theta = m\lambda \text{ (double-slit bright)}$$

$$a\sin\theta = m\lambda \text{ (single-slit dark, } m \neq 0\text{)}$$

Δy double-slit fringe spacing (m)

L slit-to-screen distance (m)

d separation of double slits (m)

a single-slit width (m)

λ wavelength (m)

When this applies

Even double-slit spacing uses the small-angle, distant-screen approximation and coherent illumination.

A method that earns marks

1. Identify whether the setup is double-slit interference or single-slit diffraction.
2. Label d as slit separation and a as slit width; they describe different distances.
3. For double-slit spacing use λL/d with all lengths in consistent units.
4. For a single slit remember that a smaller width moves the first minima outward and broadens the centre.

WORKED EXAMPLE

Doubling screen distance doubles double-slit fringe spacing. Halving a single-slit width broadens its central diffraction maximum.

1. Double-slit spacing is proportional to screen distance when wavelength and slit separation stay fixed.
2. Replacing L by 2L therefore doubles fringe spacing.
3. In the separate single-slit comparison, the first dark direction obeys sin θ = λ/a.
4. Halving a makes that angle larger, so the bright central maximum broadens.

Common mistake
The double-slit bright condition is not the single-slit bright condition.

Exam habit: Label whether you are locating a bright double-slit fringe or a dark single-slit fringe before using the order number.

SELF-CHECK

If double-slit separation doubles with screen distance and wavelength fixed, what happens to fringe spacing?

Answer: It halves.

Recognize gratings, Michelson paths and thin films

All three appear in the physical-optics section. Start by identifying which arrangement the question describes.

- A diffraction grating has many equally spaced slits. It makes sharp bright directions that separate wavelengths.
- A Michelson interferometer splits light into two arms. Each beam travels to a mirror and back, so mirror movement changes a round-trip path.
- A thin film produces interference between reflections from its two surfaces. Reflection can add a phase reversal, so do not copy the ordinary double-slit bright rule.
- For fringe counting in a simple Michelson setup, each one-wavelength change in path difference shifts the pattern by one fringe.

$\Delta\text{path} = 2\Delta x$ (one Michelson mirror moves)

In a Michelson interferometer using $500\,\text{nm}$ light in air, one mirror moves $250\,\text{nm}$ along its arm. What path change and fringe shift result?

1. Round-trip path change: $2 \times 250 = 500\,\text{nm}$.
2. This equals one wavelength.

The path changes by $500\,\text{nm}$, so one fringe passes a fixed observation point.

LESSON 10 / 12

Polarization and Malus law

After this lesson, you can

- Apply the initial half-intensity rule to unpolarized light.
- Apply Malus law using the angle between polarization and analyzer axes.

Start with: Cosine of common angles; intensity versus field amplitude; sequential fractions.

THEORY

Polarization describes the electric-field direction of a transverse light wave. An ideal polarizer transmits half the intensity of unpolarized light. An analyzer then transmits a fraction set by the angle between its axis and the incident polarization. Linear, circular and elliptical polarization describe different field-tip motions.

RULE

$$I = I_0 \cos^2 \theta$$

$$I_{\text{after first}} = I_{\text{unpolarized}}/2$$

I_0 intensity incident on the analyzer (W/m^2 or stated units)

I intensity transmitted by analyzer (same as I0)

θ angle between incident polarization and analyzer axis (degrees or radians)

When this applies

The calculation assumes ideal polarizers and initially unpolarized illumination.

A method that earns marks

1. Check whether the first incident beam is unpolarized or already linearly polarized.
2. For unpolarized light through an ideal first polarizer divide intensity by two.
3. Use that transmitted intensity as I0 for the analyzer.
4. Multiply by cos^2 θ, where θ is between the two relevant axes.

WORKED EXAMPLE

Unpolarized intensity 100 units passes a polarizer then an analyzer at 60 degrees. After the first: 50; after the second: 50 times 0.25 = 12.5 units.

1. The initial 100 intensity units are unpolarized, so the first ideal polarizer transmits 50 units.
2. The analyzer axis is at 60° to the first polarizer axis.
3. Cos 60° is 0.5 and its square is 0.25.
4. The analyzer therefore transmits 50 times 0.25 = 12.5 units, one eighth of the original intensity.

Common mistake
Do not halve already polarized intensity again before applying Malus law.

Exam habit: Square the cosine, not the angle; and do not apply an extra half factor to an already polarized analyzer input.

SELF-CHECK

Two ideal polarizers have perpendicular axes. What does the second transmit from the first?

Answer: Zero intensity in the ideal model, because $\cos^2 90° = 0$.

LESSON 11 / 12

Scattering, apparent depth and lens power

After this lesson, you can

- Calculate apparent depth near normal viewing and signed lens power.
- Explain why small-particle scattering favours shorter wavelengths.

Start with: Refraction; inverse relationships; centimetres-to-metres conversion.

THEORY

Small particles compared with wavelength produce Rayleigh scattering, which favours shorter wavelengths strongly. Droplets comparable to or larger than visible wavelengths can scatter a broad range of colours, helping explain white clouds. Refraction makes a submerged object appear shallower when viewed near the normal from air.

RULE

$$I_{\text{Rayleigh}} \propto 1/\lambda^4$$

$$d_{\text{apparent}} \approx d_{\text{real}}/n$$

$$P_{\text{lens}} = 1/f \text{ (f in metres)}$$

d_{real} actual depth (m)

d_{apparent} apparent depth (m)

n liquid refractive index relative to air approximation (dimensionless)

P lens optical power (D = m^{-1})

f signed focal length (m)

When this applies

The apparent-depth approximation assumes viewing from air near the normal; the Rayleigh rule applies to particles much smaller than wavelength.

A method that earns marks

1. Select the relevant model: small-particle scattering, near-normal depth or lens power.

2. For depth viewed from air divide real depth by the liquid index.
3. For power convert focal length to metres before taking its reciprocal.
4. Retain the focal-length sign so converging power is positive and diverging power negative.

WORKED EXAMPLE

A pool 2 m deep in water of index 1.33 appears about 1.50 m deep from air near the normal. A converging lens with f=0.25 m has power +4 dioptres.

1. For the pool, near-normal apparent depth is $2/1.33 \approx 1.50$ m, so the bottom appears shallower.
2. The index does not change the real 2 m depth; it changes the directions of the emerging rays.
3. For the separate converging lens, f = +0.25 m.
4. Its power is $1/0.25 = +4\ m^{-1}$, written +4 dioptres.

Common mistake
A focal length of 25 cm is not 25 m; using it directly gives the wrong lens power.

Exam habit: A reciprocal of a focal length in centimetres does not directly give dioptres; convert to metres first.

SELF-CHECK

A diverging lens has focal length -50 cm. What is its power?

Answer: -2 D, since -50 cm = -0.50 m.

LESSON 12 / 12

Recognize advanced optics without long derivations

After this lesson, you can

- Recognize what a quarter-wave plate changes about polarized light.
- Use Brewster angle and plane-wave relations at recognition level.

Start with: Linear and circular polarization; phase as position within an oscillation cycle; refractive index.

THEORY

The wave equation describes a disturbance propagating in space and time. Complex notation is a book-keeping method for amplitude and phase; the measured field is real. Maxwell laws connect electric and magnetic fields. Birefringent materials have different refractive behaviour for different polarizations. At Brewster incidence the reflected component can be linearly polarized.

RULE

$$\tan\theta_B = n_2/n_1$$

$$E = cB \text{ (plane wave in vacuum)}$$

θ_B Brewster incidence angle from the normal (degrees or radians)

n_1, n_2 incident and transmitted medium indices (dimensionless)

E plane-wave electric-field magnitude (V/m)

B plane-wave magnetic-field magnitude (T)

When this applies

The wave plate is operating at its intended wavelength; Brewster law here uses ordinary transparent nonmagnetic media.

A method that earns marks

1. Identify whether the question concerns phase delay, interface polarization or vacuum wave fields.
2. For a quarter-wave plate, compare the two orthogonal component amplitudes and their phase difference.
3. For Brewster incidence use the index ratio with inverse tangent.
4. For a plane vacuum wave connect field magnitudes using E = cB without a long derivation.

WORKED EXAMPLE

A quarter-wave plate can turn suitable linearly polarized light into circular polarization by introducing a quarter-cycle phase difference between equal orthogonal components. Unequal components generally give elliptical polarization.

1. Resolve suitable incident linear polarization along the plate principal axes.
2. Equal component amplitudes are required for the stated circular-polarization result.
3. A quarter-wave plate adds a quarter-cycle, or $\pi/2$, relative phase delay.
4. Equal amplitudes with this delay give circular polarization; unequal amplitudes generally give elliptical polarization.

Common mistake
These recognition notes do not replace a lecturer-assigned derivation; return to the source if such derivations are explicitly assessed.

Exam habit: For a recognition answer name the condition and consequence; deriving Maxwell equations is unnecessary for these exercises.

SELF-CHECK

Is a quarter-cycle phase delay alone enough to guarantee circular polarization?

Answer: No. The two orthogonal components must also have equal amplitudes.

Before the tests

These original questions practise the lesson objectives. They are not predictions of an exam paper. Use the separate practice book if you want all papers together.

Optics and Waves / Test A

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. Which wave can propagate through vacuum?

   A. A water wave

   B. Light

   C. A stretched-string wave

   D. Sound in air

2. With wave speed fixed, doubling frequency makes wavelength:

   A. Half as large

   B. Unchanged

   C. Four times as large

   D. Twice as large

3. Group velocity describes the motion of:

   A. The wave-packet envelope

   B. The source itself

   C. One particle only

   D. A stationary node

4. Two tones at 440 and 443 Hz produce a beat frequency of:

   A. 1320 Hz

   B. 883 Hz

   C. 3 Hz

   D. 441.5 Hz

5. At the same medium, which has greater photon energy?

   A. Lower-frequency light

   B. Both always equal

   C. Higher-frequency light

   D. Longer-wavelength light

6. Light enters glass from air. Which quantity remains unchanged?

   A. Ray direction in every case

B. Speed

C. Wavelength

D. Frequency

Part II: Show your reasoning

1. A source makes 50 complete oscillations in 10 s. Find frequency and period.

2. A wave has frequency 8 Hz and wavelength 0.75 m. Find speed. If frequency becomes 12 Hz in the same nondispersive medium, find wavelength.

3. At distance r an isotropic source has intensity 18 W per square metre. Find intensity at 3r, ignoring absorption.

4. A 1 m string fixed at both ends supports waves at 40 m/s. Find its fundamental frequency and second harmonic.

5. Light A has twice the frequency of light B. Compare photon energy and vacuum wavelength.

6. Find the critical angle for glass of index 1.5 into air of index 1.0.

Optics and Waves / Test B

30 marks | Suggested time: 30 minutes. Each MCQ: 1 mark. Each written problem: 4 marks. Attempt without notes, then review your method.

Part I: Choose one answer

1. An image with magnification -2 is:

   A. Upright and smaller

   B. Inverted and twice as tall

   C. Virtual in every case

   D. Upright and twice as tall

2. In a simple astronomical telescope, the objective usually has:

   A. A longer focal length than the eyepiece

   B. Zero focal length

   C. The same role as a polarizer

   D. No effect on the image

3. If double-slit separation increases while wavelength and screen distance stay fixed, fringe spacing:

   A. Becomes negative

   B. Increases

   C. Stays fixed

   D. Decreases

4. Two ideal crossed polarizers transmit from an initially unpolarized beam:

   A. One quarter

   B. Zero

   C. All its intensity

   D. Half

5. The unit of optical lens power is:

   A. Hertz

   B. Watt

   C. Dioptre

   D. Newton

6. A quarter-wave plate acts primarily by changing the relative:

   A. Charge of light

B. Speed of the source

C. Mass of photons

D. Phase of orthogonal components

Part II: Show your reasoning

1. A converging lens has focal length 12 cm and an object at 24 cm. Find image distance, magnification and image type.

2. A telescope objective has focal length 100 cm and eyepiece 5 cm. Find angular magnification magnitude. What happens if the eyepiece is replaced by 10 cm?

3. In a double-slit experiment, wavelength is 500 nm, screen distance 2 m and slit separation 0.50 mm. Find fringe spacing.

4. Unpolarized intensity 80 W per square metre passes through a polarizer and an analyzer at 45 degrees. Find the final intensity.

5. A diverging lens has focal length -50 cm. Find its optical power.

6. For light from air into glass of index 1.5, estimate Brewster angle.

Optics and Waves / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** Light is electromagnetic and does not require a material medium.
2. **A.** Wavelength equals speed divided by frequency.
3. **A.** The group is the envelope formed by combining nearby frequencies.
4. **C.** Beat frequency is the absolute difference.
5. **C.** Photon energy is proportional to frequency.
6. **D.** The boundary does not change the frequency of the incident oscillation.

Part II: Written solutions

1. Frequency = 50/10 = 5 Hz. Period = $1\frac{0}{5}0$ = 0.20 s. Two marks each.
2. Speed = 8 times 0.75 = 6 m/s. New wavelength = 6/12 = 0.50 m. Two marks each.
3. Inverse-square spreading gives 18/9 = 2 W per square metre. The distance triples, so the area over which power spreads increases ninefold.
4. Fundamental wavelength = 2 m; frequency = 40/2 = 20 Hz. Second harmonic = 40 Hz. Two marks per frequency with reasoning.
5. A has twice the photon energy and half the vacuum wavelength. Two marks for each relationship.
6. Sine of critical angle = $\frac{1}{1}$.5 = 0.6667, so critical angle = 41.8 degrees. Total internal reflection occurs above this angle. Credit condition, equation, value and interpretation.

Optics and Waves / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different marking scheme.

Part I: MCQs

1. **B.** The minus sign indicates inversion and magnitude 2 gives the size ratio.
2. **A.** The long-focus objective and short-focus eyepiece give a large angular-magnification ratio.
3. **D.** Fringe spacing is inversely proportional to slit separation.
4. **B.** Their axes differ by 90 degrees and $\cos^2 90^\circ$ is zero.
5. **C.** Lens power is inverse focal length in metres, expressed in dioptres.
6. **D.** A controlled phase delay changes the polarization state.

Part II: Written solutions

1. 1/di = $\frac{1}{1}$2 - 1/24 = 1/24, so di = 24 cm. m = -1. Image is real, inverted and equal size. One mark each for distance, magnification, real and inverted/equal size.
2. Initial magnitude = 10$\frac{0}{5}$ = 20. New magnitude = 100/10 = 10. Doubling eyepiece focal length halves angular magnification.
3. Convert wavelength to 0.0000005 m and separation to 0.0005 m. Spacing = (0.0000005 times 2)/0.0005 = 0.002 m = 2 mm. Credit both conversions, calculation and result.
4. First polarizer gives 40. Analyzer gives 40 times cosine squared 45 degrees = 20 W per square metre. Two marks for each stage.
5. Convert to -0.50 m. Power=1/(-0.50)=-2 dioptres. Credit conversion, formula, sign and unit.
6. Tangent of Brewster angle=1.5/1.0. The angle is arctan(1.5)=56.3 degrees. Credit correct ratio, inverse tangent, result and interpretation as incidence angle.

Source and coverage notes

The introduction and four current sections: wave motion; electromagnetic theory, photons and light; geometrical optics; physical optics. Derivations are replaced by plain-language interpretation and short exam calculations.

Source folder: Open original course folder The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. This revised study aid does not replace lecturer-specific derivations or instructions. All practice tests and suggested marking guidance are original study material.